STEMI: Primary PCI Pathway

This guideline is for adults with ST elevation myocardial infarction (STEMI) where transport to a hospital with facilities for immediate PCI can clearly occur within 90 minutes of STEMI diagnosis.

- Transmit the 12 lead ECG to the destination hospital.
- Administer 300 mg of aspirin PO.
- Confirm whether any of the following cautions to PCI are present:
  - Time of symptom onset was greater than 12 hours ago.
  - Requires assistance with their activities of daily living.
  - Severe comorbidities that significantly limit their life expectancy.
- Activate the primary PCI pathway, ensuring the following aspects are discussed if this is occurring by phone:
  - The details of any cautions to PCI that are present.
  - Confirm if ticagrelor is to be administered.
- Gain IV access, preferably in the left arm.
- Administer 180 mg of ticagrelor PO if asked to do so by the receiving clinician.
- Administer an opiate IV as required for moderate to severe pain.
- Administer 0.4 mg of GTN SL with caution and withhold GTN SL if signs of poor perfusion are present.
- If cardiac arrest occurs:
  - Commence resuscitation using the **'cardiac arrest'** guideline.
  - Administer tenecteplase IV (from the tables in the **'fibrinolysis'** guideline) if ROSC does not occur within five minutes of commencing resuscitation.
- On arrival at hospital transfer the patient directly to the PCI lab, whenever this is feasible.

Activating the primary PCI pathway

- North Shore Hospital:
  - Available if the patient will arrive in ED between 0730hrs and 1530hrs, Monday to Friday (excluding public holidays).
  - Place an early radio call to ED.
  - HHStJ personnel: Lifepak 15 transmission site codes:
    - Monday-Friday 0730-1530 NR03.

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- Auckland City Hospital:
  - Available 24/7.
  - Auckland area: place an early radio call to ED if the patient will arrive in ED between 0730hrs and 1530hrs, Monday to Friday. Outside of these hours phone the interventional cardiologist on 0800 366 666.
  - Northland area: phone the interventional cardiologist on 0800 366 666.
  - HHStJ personnel: Lifepak 15 transmission site code NR01.
- Middlemore Hospital:
  - Available if the patient will arrive in ED between 0730hrs and 1530hrs, Monday to Friday (excluding public holidays).
  - Place an early radio call to ED.
  - HHStJ personnel: Lifepak 15 transmission site codes:
    - Monday-Friday 0730-1530 NR02.
- Waikato Hospital:
  - Available 24/7.
  - Phone the on call cardiologist on 07 839 8676.
  - HHStJ personnel: Lifepak 15 transmission site code CR01 or NR07.
- Tauranga Hospital:
  - Available if the patient will arrive in ED between 0730hrs and 1530hrs Monday to Friday (excluding public holidays).
  - Phone the cardiac catheter lab coordinator on 027 7047712.
  - HHStJ personnel: Lifepak 15 transmission site code CR04.
- Wellington Hospital
  - Available 24/7.
  - WFA personnel: phone the Cardiology Nurse Coordinator on 021 504746.
  - All other personnel: call the on call doctor for HHStJ via the Clinical Desk.
  - HHStJ personnel: Lifepak 15 transmission site code CLINICAL DESK.
- Nelson Hospital:
  - Available if the patient will arrive in ED between 0800hrs and 1600hrs, Monday to Friday (excluding public holidays).
  - Call the on call cardiologist on 0800 728 4929, extension 2.
  - HHStJ personnel: Lifepak 15 transmission site code SI02.

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- Christchurch Hospital:
  - Available 24/7.
  - Call the on call cardiology registrar on 0800 728 4929, extension 1.
  - HHStJ personnel: Lifepak 15 transmission site code SI01.
- Dunedin Hospital:
  - Available 24/7.
  - Call the on call cardiologist on 0800 728 4929, extension 3.
  - HHStJ personnel: Lifepak 15 transmission site code SI03.
- Seek clinical advice if contact is not able to be made with the clinician listed above.

Backup

- If a Paramedic is present, backup from an ICP/CCP is not routinely required.
- EMTs must request backup from a Paramedic or ICP/CCP.
- Backup from an ICP/CCP should be requested if the patient has:
  - Bradydysrhythmia, or
  - Tachydysrhythmia, or
  - Signs of shock.

Additional information

General principles

- Minimise scene time and avoid unnecessary delays as outcomes are primarily dependent on how quickly the occluded coronary artery is opened.
- Primary PCI is the preferred reperfusion pathway provided the patient can clearly be transported to a hospital with the facilities for immediate PCI within 90 minutes of the diagnosis of STEMI. However, the balance of risk is in favour of administering fibrinolysis if there will be any delays preventing transport being achieved within 90 minutes.
- Be prepared to treat cardiac arrest but do not routinely attach defibrillation pads unless the patient is being transported by helicopter. Cardiac arrest will occur in approximately 5% of patients, the initial rhythm will usually be ventricular fibrillation, and most patients will achieve return of spontaneous circulation after 1-2 shocks.

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- IV access is preferably gained in the left arm because the interventional cardiologist will usually gain access via the right radial artery.
- Have a low threshold for administering antiemetics.
- Severe comorbidities that significantly limit life expectancy include dementia, severe COPD, severe heart failure, a requirement for aged residential care and metastatic cancer with weight loss.

Response and/or transport by helicopter

- Response by helicopter may be indicated if a CCP is required for the management of a dysrhythmia or cardiogenic shock, and helicopter arrival at scene is at least 15 minutes faster than a road response.
- Transport by helicopter may be indicated if the patient is following the primary PCI pathway, and it will save more than 15 minutes compared to road transport.
- For helicopter transport criteria to be met, the road transport time will usually need to be more than 75 minutes.

More Information

Version history

11 September 2023 1.0.4

New: Guideline released.